

NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:

- 1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (WOLFF3D_OmniKernel_Clevis_Standard_25x25.STEP) DEFINES NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCE: LINEAR +/- 0.10 mm, ANGULAR +/- 0.5 DEG (PER ISO 2768-m).

2. MATERIAL & ADDITIVE MANUFACTURING:

- 2.1 MATERIAL: ALUMINUM ALLOY AISi10Mg CONFORMING TO ASTM F3318 / AMS 4880.
2.2 FABRICATION PROCESS: LASER POWDER BED FUSION (PBF-LB / SLM).
2.3 LAYER THICKNESS: 30 TO 60 MICRONS.
2.4 VIRGIN/RECYCLED POWDER RATIO AND SPHERICITY SHALL SATISFY AMS 7000 SPECIFICATIONS.

3. THERMAL PROCESSING & POWDER EVACUATION:

- 3.1 STRESS RELIEF ANNEAL PRIOR TO BASEPLATE CUTTING: 300°C +/- 10°C FOR 2 HOURS, AIR COOL.
3.2 100% UNMELTED POWDER SHALL BE REMOVED THROUGH 2X DIA 4.00 mm DEPOWDER PORTS VIA ULTRASONIC VIBRATION AND DRY NITROGEN FLUSHING.
3.3 RESIDUAL POWDER VERIFICATION VIA GRAVIMETRIC DRIFT (< 0.15 g) OR RADIOGRAPHIC TESTING.
3.4 SEAL PORTS WITH DIA 6.5 mm FLUSH PLUGS RETAINED VIA LOCTITE 243 OR LASER MICRO-WELDING.

4. MACHINING & THREADING (POST HEAT-TREATMENT):

- 4.1 BASE MOUNTING FLANGE (DATUM A) SHALL BE CNC FINISH-MILLED TO ENSURE PLANARITY.
4.2 CRITICAL PIN BORE (DIA 10.000 H7) SHALL BE PRECISION REAMED; LEAD-IN CHAMFERS 0.8 mm X 45 DEG REQUIRED ON ENTRY/EXIT FACES.
4.3 4X MOUNTING HOLES SHALL BE TAPPED TO M5-6H THRU (PRE-DRILLED DIA 4.20 mm AS PRINTED).
4.4 REMOVE ALL BURRS AND BREAK SHARP EDGES 0.1 - 0.3 mm; MAINTAIN R3.0 mm FILLETS AS MODELLED.

5. SURFACE FINISH & PROTECTIVE COATING:

- 5.1 AS-PRINTED EXTERNAL SURFACES: GLASS BEAD BLASTING TO Ra 3.2 - 6.3 MICRONS (PER ASTM F3301).
5.2 CNC MACHINED SURFACES (DATUM A & DIA 10 BORE): Ra <= 0.8 MICRONS.
5.3 SURFACE FINISH: CHEMICAL CONVERSION COATING PER MIL-DTL-5541 TYPE II CLASS 1A (ALTERNATIVE: HARD ANODIZE PER MIL-A-8625 TYPE III CLASS 1, COATING THICKNESS 25-35 MICRONS).

6. NON-DESTRUCTIVE TESTING (NDT) & QUALITY ASSURANCE:

- 6.1 100% VISUAL AND FLUORESCENT PENETRANT INSPECTION (FPI) PER ASTM E1417 (ACCEPTANCE CLASS A).
6.2 INTERNAL CONFORMAL LATTICE AND BOUNDARY FUSION INTEGRITY SHALL BE QUALIFIED VIA COMPUTED TOMOGRAPHY (CT) SCAN PER ASTM E1441.

